Efficient Visualization of Knowledge Bases via Space-Time Graphs: The OHARA Architecture

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***Abstract*—Flat-chunk retrieval-augmented generation (RAG) returns ranked lists that give users no spatial or temporal model of a growing corpus. We present OHARA, an architecture built on a *Space-Time Graph* that unifies document structural hierarchy, temporal decay, and cross-document entity/ontology pivots as a single substrate for both retrieval and a novel 3D sunburst-tunnel visualization (time $\rightarrow$ Z-axis, ontology $\rightarrow$ sunburst cross-section, document structure $\rightarrow$ radial discs). An eight-phase hybrid retrieval engine fuses lexical, dense, ontology, entity, structural, and cross-document signals into tiered, provenance-carrying results. On two corpora (QASPER-200 academic papers, MultiHop-RAG-609 news articles) the tuned fusion pipeline reaches ranking *parity* with the best single dense signal, not superiority; the differentiated, quantitatively supported result is a corroboration-gated Principal tier that abstains on 45.6% of unanswerable queries versus 0.0% for a plain top-k cut, at 91.5% Principal-hit on answerable queries. Ingest builds the full semantic graph at roughly $12 per 1,000 documents; the visualization renders sub-linearly to 12,323 nodes. We report these results honestly, including a negative finding on temporal decay scoring, and frame OHARA's contribution as a knowledge base that is *seeable and auditable* rather than one that outranks dense retrieval.**

***Index Terms*—retrieval-augmented generation, knowledge graph, information visualization, ontology, temporal reasoning**

I. Introduction

Flat-chunk RAG pipelines return a linear list of passages; users lose the spatial or temporal context of where each passage sits in the corpus, compounding the "lost in the middle" effect of long contexts [1]. OHARA targets corpora that carry, or can be given, structural metadata (sections, paragraphs, tables) and publication dates – regulatory filings, academic papers, technical manuals, long-form journalism.

Contributions. (a) An efficient 3D sunburst-tunnel visualization mapping time to the Z-axis, ontology to a sunburst polar plane, document structure to radial discs, and decay class to a temporal-reach aura. (b) A Space-Time Graph data model combining structural hierarchy, temporal decay, and cross-document entity pivots, serving as both retrieval substrate and visualization coordinate system. (c) A multi-phase hybrid retrieval engine whose measured value is tiered, provenance-carrying results at parity with the best single-signal retriever, not ranking gains. (d) A corroboration-gated Principal tier that abstains on unanswerable queries where a top-k cut structurally cannot – at identical retrieval, the $\geq$2-signal gate abstains on 45.6% of MultiHop-RAG null queries versus 0.0% for plain top-k, holding 91.5% Principal-hit on answerable queries. (e) A SUMO ontology-grounded semantic layer that anchors the visualization's coordinates and enables tag-expansion retrieval.

TABLE I
Capability comparison against graph-RAG baselines.

Capability	GraphRAG	LightRAG	HippoRAG	OHARA
Entity/relation subgraph	✓	✓	✓	✓
Structural hierarchy (DoCO)	–	–	–	✓
Cross-doc similarity edges	partial	✓	✓	✓
SUMO ontology grounding	–	–	–	✓
Temporal decay scoring	–	–	–	✓
3D space-time visualization	–	–	–	✓
Tiered explainability	–	–	–	✓

We state our positioning explicitly: OHARA does not claim graph-augmented retrieval outranks dense retrieval – our evaluation (Sec. VII) shows the tuned hybrid *matches* the best single signal. The thesis is that a knowledge base should be seeable and auditable: the same graph that answers a query renders as a navigable spatial-temporal map and attaches per-result provenance, abstaining rather than emitting k confident-looking passages when nothing is corroborated.

II. Related Work

Graph-based RAG systems (GraphRAG, LightRAG, HippoRAG) build entity/relation subgraphs but discard document structural hierarchy and couple retrieval to no temporal or spatial model. Explainable-retrieval work (Self-RAG, Corrective RAG) targets provenance but not 3D navigation. Prior 3D and temporal graph visualizations (Time-arc, Storyline, VR graph tools) do not combine an ontology sunburst cross-section with a temporal tunnel axis; OHARA's layout builds on the 2D radial space-filling hierarchy tradition [2] extended into a temporal third dimension so topic columns remain traceable through time. Table I summarizes the gap: no compared system combines structural hierarchy, SUMO ontology grounding, temporal decay, and tiered explainability.

III. The Space-Time Graph Data Model

The Space-Time Graph is a 5-tuple $G = (V, E, \tau, \delta, \sigma)$. The vertex set $V = V_{docs} \cup V_{sect} \cup V_{para} \cup V_{tab} \cup V_{ent}$ spans document roots, structural hierarchy nodes, content nodes (paragraphs/tables), and named entities. Edges $E \subseteq V \times V \times R$ are typed by seven relations

$$R = \{\text{HAS\_CHILD}, \text{NEXT\_SIBLING}, \text{BELONGS\_TO}, \text{MENTIONS}, \text{RELATED\_TO}, \text{SIMILAR\_TO}, \text{TOC\_REF}\},$$

spanning structural (tree), semantic (cross-cutting), and cross-document (Jaccard-gated) relationships. Temporal bucketing $\tau : V_{docs} \rightarrow \Theta$ maps documents to a Z-axis coordinate at configurable resolution; decay classification $\delta : V_{docs} \rightarrow D$ assigns one of four classes (EVERGREEN λ=10^{-6}, SCHOLARLY λ=10^{-4}, CURRENT λ=10^{-2}, EPHEMERAL λ=10^{-1}), auto-promoted to EVERGREEN when in-degree of SIMILAR_TO ≥ 5; narrative enrichment σ attaches a verb, tags, summary, and temporal relation to cross-document edges. Decay scoring uses $w \cdot e^{-\lambda \Delta t}$ under a five-layer protection scheme (no temporal intent, Principal-tier immunity, high-BM25 immunity, weak-candidate-only, and date-range coverage boost) that we present as a guard-railed recency prior rather than a demonstrated ranking gain: on the one temporal benchmark evaluated the prior is neutral-to-slightly-negative (Sec. VII, finding 3), with genuine freshness domains (news monitoring, versioned documentation) left as future validation.

Pre-filtering complexity. At ingest, each document's paragraph-level `entity_slugs` and `sumo_tags` are unioned onto its root node (document rollup). Query-time candidate pruning then scans $O(n)$ document rollups, $n = |V_{docs}|$, instead of $O(n \times \text{chunks\_per\_doc})$ content nodes – a ∼50–60× reduction at our measured corpus densities (16 sections, 44 paragraphs per document). Whether the subsequent *descent* into the pruned document finds the correct paragraph is the empirical bottleneck, not the pre-filter's cost: our evaluation attributes ∼42% of corpus-wide retrieval failures to document location and descent (Sec. VII-A).

IV. Multi-Phase Hybrid Retrieval Engine

Eight signal phases run in a configurable pipeline: (0) query fingerprinting – keyword/phrase/paragraph classification with Gemini-extracted SUMO tags, entity hints, and temporal intent for phrase-length queries and above; (0b) TOC-guided section selection, an LLM pass over seed document tables of contents that supplies structural traversal entry points; (1) BM25 full-text; (1b) SUMO tag overlap with hierarchy expansion; (1c) multi-hop SIMILAR_TO traversal carrying narrative edge verbs and summaries; (1d) dense vector similarity (`gemini-embedding-2`, 768-d, cosine); (2) entity pivot across shared canonical entities; (3) structural traversal (depth 2) with a corrective filter dropping zero-SUMO-overlap nodes. Phase 4 fuses per-phase max-normalized scores with a weighted sum (base weights BM25 1.0, vector 0.5, SUMO 0.4, entity 0.6, cross-doc 0.4, structural 0.3, scaled by query-mode multipliers) and adds the guarded temporal contribution.

Tier classification. Grounded in Bates' Berrypicking and Pirolli's Information Foraging Theory: *Principal* nodes require $\geq$2 independent phase contributions, cross-document or multi-document evidence, and score above the 75th percentile; *Integrity* adds verified structural and cross-document neighbors; *Explorer* is a metadata-only frontier one hop beyond Integrity, filtered to a weakening-scent weight band. In practice the corroborating pair for Principal-tier membership is almost always `bm25∩vector` – the graph-side phases (entity, cross-document, structural) exclude already-seen nodes by design and so expand the frontier rather than corroborate the core. This constraint's quantitatively confirmed payoff is abstention, not ranking (Sec. VII, finding 2): a top-k cut cannot express "no confident answer," while the $\geq$2-signal Principal gate can and does.

V. Ingest Pipeline

Parsing (LiteParse for PDF/EPUB/DOCX, Markdown pass-through) is followed by content-hash-cached LLM structuring into DoCO node types (`gemini-2.5-flash-lite`, concurrency 4), three-stage SUMO tag validation against a 22,700-entry index (exact, case-insensitive, alias table), entity extraction with canonical cross-document deduplication, structural/semantic edge creation with document rollup, and Jaccard-gated cross-document similarity enriched with an LLM-generated verb, tag set, summary, and temporal relation per edge. Measured at scale, ingesting 200 QASPER papers consumed 6.2M prompt + 4.5M output tokens (∼53k/doc), \$2.41 at standard rates (∼\$1.20 flex tier) – roughly \$12 per 1,000 documents for the full semantic graph. Content-hash caching makes idempotent re-ingest near-free: 14.5% of documents suffered a transient API failure on first pass, all recovered at near-zero marginal cost. One caveat: forced re-ingest of partial documents duplicated structural edges (23k duplicates across 13k pairs), so ingest idempotency currently holds at chunk level, not edge level.

VI. Space-Time Graph Visualization

The visualization rests on three irreducible dimensions no 2D projection preserves simultaneously: time, ontology, and structure. The Z-axis carries resolution-bucketed publication time; the XY cross-section is an ontology sunburst plane in which each document sits at the polar coordinate of its top SUMO category (θ from a precomputed recursive angular partition of the SUMO tree, $r = R_{max}\sqrt{depth/D_{max}}$), so documents sharing a category stack into a stable topic column traceable along Z. Each document unfolds locally as a disc perpendicular to Z: the document node at center, section nodes in concentric rings by hierarchy level, paragraphs and tables on the outermost ring (Fig. 1). Named entities sit at the centroid of the documents that mention them, pushed outward by a fixed offset; semi-transparent decay-class auras (5/2.5/1/0.4 disc-spans for EVERGREEN/SCHOLARLY/CURRENT/EPHEMERAL) make temporal reach spatially legible even where decay does not help ranking. Rendering uses Three.js `InstancedMesh` batching (one draw call per shape bucket) with lazy per-document, per-node loading. Measured on QASPER (headless Chromium, software WebGL, a conservative lower bound):

TABLE II
QASPER RETRIEVAL QUALITY (150 QUESTIONS, CORPUS-WIDE OVER 229 DOCS).

Config	H@4	H@10	MRR@10	P-hit
BM25-only	12.7%	25.3%	0.102	0.0%
Vector-only	**26.7%**	**33.3%**	0.175	**24.7%**
Full (default)	22.0%	30.0%	0.164	22.7%
Full (tuned)	25.3%	**33.3%**	**0.179**	22.7%

TABLE III
MULTIHOP-RAG RETRIEVAL QUALITY (500 STRATIFIED QUERIES).

Config	H@10	MRR@10	P-hit	Null abst.
BM25-only	94.4%	0.727	4.3%	96.8%*
Vector-only	**98.4%**	0.811	**92.5%**	13.6%
Full (default)	96.5%	0.791	91.5%	**45.6%**
− Corrob.	96.8%	0.795	92.0%†	0.0%
Full (tuned)	**98.4%**	**0.812**	92.0%	31.2%

*Degenerate: near-empty Principal tier, not discrimination. †Proxied as top-5.

scene rebuild and JS heap grow sub-linearly from 713 to 12,323 nodes (1.5 s→2.3 s rebuild, 52 MB→80 MB heap, a 17× node increase yielding only ∼1.7×/∼1.5× growth), confirming the practical ceiling is visual clutter and fill-rate, not geometry submission. A within-subjects user study protocol (flat list vs. sunburst-tunnel, $n \geq 8$, temporal filtering / topic tracing / structural lookup tasks) is fully specified but not yet run.

VII. EVALUATION

We evaluate on two corpora: QASPER (200 academic papers, 150 answerable questions, paragraph-level gold, scored corpus-wide over 229 documents including distractors) and MultiHop-RAG (609 news articles, 500 stratified queries including 125 null/unanswerable, document-level gold). A preliminary smoke run surfaced two silent failure modes fixed before the reported runs: raw BM25 magnitudes (∼13–24) dominating bounded phase contributions absent per-phase normalization, and a stored/query-side embedding-model mismatch; both are corrected in all figures below.

A. QASPER Retrieval Quality

Table II reports the headline configurations. BM25-only badly underperforms vector-only (25.3% vs. 33.3% Hits@10) because corpus-wide lexical overlap collides across 229 documents – the inverse of BM25's role *within* one paper, where it is the strongest signal. The default-weight full pipeline (30.0% Hits@10) trails vector-only because BM25 at weight 1.0 ranks lexical noise above semantic hits on this paraphrase-heavy benchmark; a grid search over (BM25, vector) weights on a held-out subset selects (0.6, 1.0), which recovers the gap entirely and slightly exceeds vector-only on MRR (0.179 vs. 0.175). Decomposing failures against an oracle that ranks only within the gold document shows corpus-wide loss concentrates in document *location*: ∼42% of queries fail before the correct document is even reached, while paragraph discrimination given the correct document succeeds for roughly two-thirds of queries (oracle Hits@10 64.4%, in the band of published within-document baselines).

B. MultiHop-RAG Retrieval Quality and Abstention

The QASPER-tuned weights (0.6, 1.0) transfer unchanged to MultiHop-RAG (Table III), reaching Hits@10/MRR parity with vector-only while exceeding it on MAP (0.556 vs. 0.538). Three findings decide the paper's central claims. **First**, weight transfer generalizes: the lexical-first *default* prior, not the fusion architecture, was the gap on both corpora. **Second**, corroboration-gated abstention is the one differentiated quantitative win: comparing full vs. the same pipeline with the corroboration constraint removed (identical retrieval, differing only in the tier rule) shows the constrained Principal tier abstains on 45.6% of unanswerable queries versus 0.0% for a plain top-k cut, at 91.5% Principal-hit on answerable queries; tuning trades some of this away (31.2%) as an explicit precision/abstention trade-off. **Third**, temporal decay is an honest negative here: removing it *improves* MRR on the temporal query slice (0.749 vs. 0.736), because MultiHop-RAG's temporal queries test event ordering, not recency, and the guard layers bound the resulting damage to ∼1pp. Per-phase ablations on both corpora land within ±0.5pp of the full pipeline – graph phases (SUMO, cross-document, structural, TOC) contribute provenance and abstention, not ranking lift, on the corpora tested. An end-to-end generation check (top-10 context, `gemini-2.5-flash-lite`) shows 54% (tuned) vs. 57% (vector-only) accuracy, mirroring retrieval parity; the dominant failure (41%) is insufficient retrieved context, not hallucination.

VIII. DISCUSSION

Limitations. Single LLM backend (Gemini) throughout structuring, tagging, and enrichment. Both evaluation corpora are clean born-digital text; noisy OCR and non-English corpora are untested. Fusion weights were tuned on one QASPER subset and, while shown to transfer to MultiHop-RAG, two corpora do not establish domain-general defaults. Visualization rendering is validated to 12k nodes but grows visually cluttered past ∼50 selected documents; the user study protocol is prepared but not yet run.

Research agenda. We separate measured results from open hypotheses. (H1) Ranking gains from spatial-temporal structure: today's parity result caps corroboration at two signal families because graph phases exclude already-seen nodes; letting them re-score seen nodes to enable many-angle corroboration is the direct next experiment. (H2) Recency-semantic temporal scoring: today's result is neutral-to-negative on event-ordering queries; corpora with genuine freshness semantics (news monitoring, versioned documentation) remain untested. (H3) The human navigation benefit of the 3D spatial-temporal mental model is the thesis claim, untested by the scope of this paper, with protocol and rendering substrate both validated and ready.

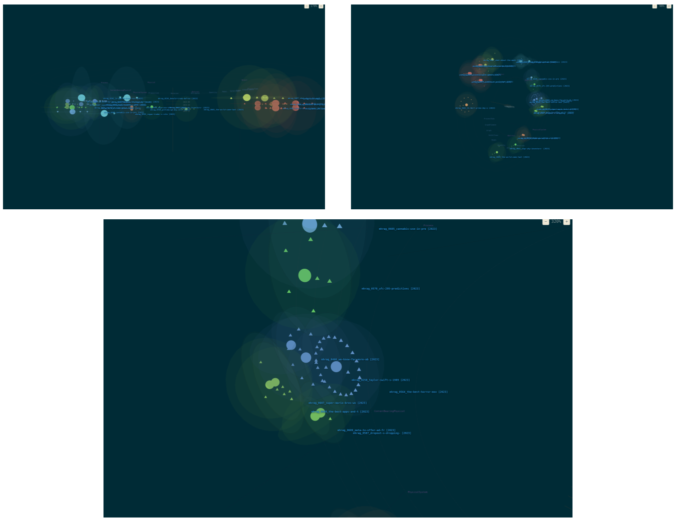

Fig. 1. Three camera views of the same 25-document Space-Time Graph (color = top SUMO category per document). *Top left*: sunburst top-down, looking down $-$Y along the ontology cross-section – documents cluster into SUMO category columns (*Process*, *Physical*, *Object*, *Proposition* labels visible). *Top right*: oblique tunnel view, camera offset along Z so publication time reads as depth. *Bottom*: close-up on a single document disc, showing its section hierarchy as a concentric ring of nodes (triangles) around the document root (sphere).

IX. Conclusion

OHARA's Space-Time Graph – structural hierarchy, temporal decay, and cross-document ontology unified in one substrate – renders as a navigable 3D sunburst-tunnel and feeds a multi-phase retriever that delivers tiered, provenance-carrying results at parity with the best single-signal retriever, not superiority over it. The differentiated, quantitatively supported contribution is corroboration-gated abstention: a knowledge base that says "I don't know" on 45.6% of unanswerable queries where a top-k cut cannot, at no cost to Principal-hit rate on answerable ones. Ingest builds the full semantic graph at $\sim$\$12 per 1,000 documents; rendering scales sub-linearly. We frame this work as the opening of a research line: the substrate, harness, and honest "not-yet" results (Sec. VII) are the foundation for the open hypotheses above.

Acknowledgments

This research is supported by research funding from University of Science, Vietnam National University – Ho Chi Minh City.

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